Yocto/Buildroot Comparison

Advanced Embedded Linux Development

with **Dan Walkes**



University of Colorado **Boulder**

Learning objectives:

Comparison of Yocto and Buildroot build frameworks

That was too complicated!

- Why not just cross compile the binary yourself and copy it into the rootfs image?
  - Hardware changes are easier to manage.
  - Emulation is easier to use with a build framework.
  - Hacks become unmanageable long term.
  - Technical Debt is a recurring cost.

Which is Better? Buildroot or Yocto?

- This is a question of religion!
- There's no "right" answer
- Relates to preferences and possibly personal bias
- I will share my preferences and bias!



Yocto and Buildroot Similarities

- Both have really talented and responsive developers.
- Both are well thought out and well maintained.
- Both have similar support for common embedded platforms (x86, ARM).
- Both have extensive and well written documentation.

Yocto and Buildroot Differences

Criteria	Buildroot	Yocto
Simplicity / Initial Learning Curve	Simplest	More Complex
Developer workflow support	Good	Best
Clean Build Time	Shorter	Longer
Incremental Build Support	Moderate	Very Good
Platform/Package support	Good	Best

When Would I Consider Buildroot?

- A project already using it!
- A team already familiar with it.
- A team with lots of Makefile/Linux Kernel build experience and relatively little Python/Yocto skills.
- A team that's **not excited** about learning Yocto.
- A platform well supported on Buildroot (especially a mature platform).
- A product with few and simple requirements.
- A product with a well defined roadmap.

When Would I Consider Yocto?

- A product with more complex requirements/more undefined roadmap.
- A newer hardware platform
  - Especially if not supported yet on Buildroot or the manufacturer has not committed to Buildroot support.
- A team excited about the challenge of learning Yocto!